MODEL LAB · 2026-10-27

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

17,710
held-out plate appearances

2.92%
held-out HR rate

0.6014
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-13, scored on the 17,710 that came after (actual HR rate 2.92%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5820	1.61×	0.02829	0.13108	0.00466
KNN comps Prior is the pooled rate of the nearest comparables.	0.6014	1.64×	0.02824	0.12993	0.00614
SVR Support-vector regression on the same feature matrix.	0.5940	1.51×	0.02829	0.13054	0.00689
Random forest Forest regression; a learned similarity metric.	0.5967	1.51×	0.02828	0.13022	0.00676
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6014	1.64×	0.02824	0.12993	0.00614
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6014	1.64×	0.02824	0.12993	0.00614
Linear regression Ordinary least squares on the same features.	0.5979	1.57×	0.02828	0.13031	0.00488
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5969	1.59×	0.02833	0.13060	0.00615
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5971	1.55×	0.02827	0.13021	0.00494
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5961	1.45×	0.02832	0.13060	0.00718

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **260** knn **260** svr **260** rf **260** core **260** ensemble **260** linear **260** logistic **260** nn **260** xgb **260**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (1,620 PA, actual HR rate 2.35%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.6456	1.58×	0.02291	0.11126	0.01790
KNN comps Prior is the pooled rate of the nearest comparables.	0.6062	1.58×	0.02288	0.10980	0.01007
SVR Support-vector regression on the same feature matrix.	0.5783	1.58×	0.02323	0.11613	0.02309
Random forest Forest regression; a learned similarity metric.	0.6000	1.32×	0.02320	0.11319	0.02260
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6062	1.58×	0.02288	0.10980	0.01007
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6062	1.58×	0.02288	0.10980	0.01007
Linear regression Ordinary least squares on the same features.	0.6112	1.32×	0.02322	0.11383	0.02025
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6015	1.32×	0.02354	0.11442	0.02384
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.58×	0.02300	0.11128	0.01557
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5934	1.05×	0.02361	0.11640	0.02762

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 17.0%, Junior Caminero 16.4%, William Contreras 15.7%, Jake Bauers 15.1%, Jonathan Aranda 13.1%, Brice Turang 13.1%
Flat league	5/6	Jackson Chourio 17.3%, Junior Caminero 15.7%, William Contreras 15.6%, Jake Bauers 14.0%, Brice Turang 13.3%, Yandy Díaz 13.2%
KNN comps	5/6	Jackson Chourio 17.6%, Junior Caminero 16.2%, William Contreras 15.7%, Jake Bauers 15.2%, Jonathan Aranda 13.2%, Yandy Díaz 13.0%
SVR	5/6	Jackson Chourio 17.9%, Junior Caminero 17.6%, William Contreras 15.7%, Jake Bauers 15.0%, Yandy Díaz 13.4%, Brice Turang 13.0%
Random forest	5/6	Jackson Chourio 17.8%, Junior Caminero 16.9%, William Contreras 15.6%, Jake Bauers 15.3%, Yandy Díaz 13.3%, Jonathan Aranda 12.8%
Core (KNN+SVR)	5/6	Jackson Chourio 17.6%, Junior Caminero 16.2%, William Contreras 15.7%, Jake Bauers 15.2%, Jonathan Aranda 13.2%, Yandy Díaz 13.0%
Forest blend	5/6	Jackson Chourio 17.6%, Junior Caminero 16.2%, William Contreras 15.7%, Jake Bauers 15.2%, Jonathan Aranda 13.2%, Yandy Díaz 13.0%
Linear regression	5/6	Jackson Chourio 17.8%, Junior Caminero 16.4%, William Contreras 15.7%, Jake Bauers 15.0%, Jonathan Aranda 13.1%, Yandy Díaz 13.1%
Logistic regression	5/6	Jackson Chourio 17.8%, Junior Caminero 16.9%, William Contreras 15.7%, Jake Bauers 15.0%, Yandy Díaz 13.0%, Jonathan Aranda 12.9%
Neural net (torch)	5/6	Jackson Chourio 17.8%, Junior Caminero 16.8%, William Contreras 15.8%, Jake Bauers 15.2%, Yandy Díaz 13.0%, Jonathan Aranda 13.0%
XGBoost	5/6	Jackson Chourio 18.0%, Junior Caminero 17.1%, William Contreras 15.6%, Jake Bauers 15.3%, Yandy Díaz 13.4%, Jonathan Aranda 12.9%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jackson Chourio**	MIL	**17.0%**	17.3%	17.6%	17.9%	17.8%	17.6%	17.6%	17.8%	17.8%	
2	**Junior Caminero**	TB	**16.4%**	15.7%	16.2%	17.6%	16.9%	16.2%	16.2%	16.4%	16.9%	
3	**William Contreras**	MIL	**15.7%**	15.6%	15.7%	15.7%	15.6%	15.7%	15.7%	15.7%	15.7%	
4	**Jake Bauers**	MIL	**15.1%**	14.0%	15.2%	15.0%	15.3%	15.2%	15.2%	15.0%	15.0%	
5	**Jonathan Aranda**	TB	**13.1%**	12.5%	13.2%	12.7%	12.8%	13.2%	13.2%	13.1%	12.9%	
6	**Brice Turang**	MIL	**13.1%**	13.3%	12.8%	13.0%	12.6%	12.8%	12.8%	12.3%	12.2%	
7	**Yandy Díaz**	TB	**12.9%**	13.2%	13.0%	13.4%	13.3%	13.0%	13.0%	13.1%	13.0%	
8	**Christian Yelich**	MIL	**12.1%**	12.1%	12.4%	11.8%	12.1%	12.4%	12.4%	12.3%	11.8%	
9	**Garrett Mitchell**	MIL	**12.0%**	11.5%	11.9%	11.1%	11.5%	11.9%	11.9%	11.4%	11.4%	
10	**Ryan Vilade**	TB	**11.3%**	10.7%	11.2%	10.9%	11.3%	11.2%	11.2%	10.9%	10.7%	
11	**Jonny DeLuca**	TB	**8.8%**	9.6%	8.4%	8.4%	8.1%	8.4%	8.4%	8.5%	8.8%	
12	**Liam Hicks**	TB	**8.0%**	8.9%	7.9%	8.6%	8.5%	7.9%	7.9%	8.3%	8.1%	
13	**Cooper Pratt**	MIL	**7.0%**	9.0%	6.9%	6.6%	6.5%	6.9%	6.9%	5.9%	6.3%	
14	**David Hamilton**	MIL	**6.6%**	7.9%	6.4%	6.0%	5.8%	6.4%	6.4%	6.3%	6.3%	
15	**Sal Frelick**	MIL	**6.5%**	8.2%	6.7%	6.2%	6.0%	6.7%	6.7%	6.5%	6.6%	
16	**Richie Palacios**	TB	**6.2%**	6.9%	6.1%	5.9%	5.4%	6.1%	6.1%	5.8%	5.9%	
17	**Taylor Walls**	TB	**4.3%**	5.6%	4.1%	3.2%	3.2%	4.1%	4.1%	4.0%	4.1%	
18	**Chandler Simpson**	TB	**2.3%**	5.2%	3.1%	1.4%	1.6%	3.1%	3.1%	1.3%	2.7%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

Slate for 2026-10-27 · model 2026.10.01-stack · built 2026-10-27 20:04:46 · live data 2026-10-27 20:04:46

Built for research and entertainment. Probabilities are estimates, not guarantees; never stake what you cannot afford to lose. Data: MLB Stats API play-by-play, lineups and live scores; ESPN injuries and bios.